

Himanshu Sharma

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EXPERIENCE

Amazon (Ads, Applied Science)

Bengaluru, India

Applied Scientist

Jan 2024 – Present

- Designed a **retrieval-augmented classification engine** (a LoRA-tuned vision-language retriever distilled from a larger teacher, paired with an LLM over retrieved neighbours), onboarding new policies and labels without per-label retraining.
- Led a **moderation efficiency initiative** replacing many policy-specific models with one unified model (custom XLM-RoBERTa + ConvNeXtV2), delivering **~20% automation** over **~13M weekly ads**; owned the roadmap, the AWS CDK deployment pipelines, and cross-team rollout.
- Designed and launched a multimodal, multilingual, multilabel **ad moderation model**, pretraining and finetuning **FLAVA** and **VLMO** to **>90% volume-weighted precision** across global marketplaces.
- Restored continuous delivery across the team's ML pipelines as owner of a team-wide **operational excellence** initiative, raising the share shipping reliably to production from **15% to 76%**; set the health bar, and built a repeatable remediation and mechanism adopted by every pipeline owner.
- Built an **agentic RAG** knowledge system on AWS Bedrock Knowledge Bases; **runner-up, AT Hackathon 2024** for a model-monitoring and drift-detection framework.

RBCCPS & ARTPARK, Indian Institute of Science

Bengaluru, India

Technical Associate / Consultant

Jul 2023 – Dec 2023

- Fused LiDAR and depth-camera measurements with Kalman filtering for autonomous-mobile-robot localisation, and implemented multi-robot map merging in ROS to combine distributed SLAM outputs into one global map.

EDUCATION

Indian Institute of Science

Bengaluru, India

M.Tech (Res) in Electrical Communication Engineering; CGPA: 8.3/10

Oct 2020 – Apr 2023

Rajasthan Technical University

Kota, India

B.Tech in Electronics & Communication Engineering; CGPA: 8.08/10

Jul 2015 – Jun 2019

PUBLICATIONS

- “Ordinal Classification with Risk Guarantee: From Confidence to Future Assurance”, *Amazon Machine Learning Conference*, 2025: conformal risk control for asymmetric-cost ordinal moderation; **97.4% precision** under finite-sample guarantees
- “InSight: Maximizing Impact with Minimal Labeling”, *Amazon Machine Learning Conference*, 2025: influence-based active learning; **36% fewer labels**, **20% moderator productivity** in production
- “Vision-driven Tele-Operation for Robot Arm Manipulation”, *7th International Conference on Robotics & Automation Engineering (ICRAE)*, Singapore, 2022

SELECTED PROJECTS

Language-Conditioned Vision-Language-Action Policy (Flow Matching) | *PyTorch, MuJoCo*

- Built a **flow-matching** action expert (**99.9M** trainable params) that cross-attends to a frozen **SmolVLM2** vision-language backbone: linear noise-to-action interpolation, velocity-field regression, and few-step Euler integration at inference, conditioned on free-form natural-language task instructions.

Diffusion Policy for Robotic Manipulation | *PyTorch, MuJoCo, SageMaker*

- 79.9M-parameter** conditional DDPM over action trajectories built with **diffusers**: cosine β schedule, 10-step **DDIM**, **1-D temporal U-Net** with **FiLM**, SpatialSoftmax encoder, EMA. Benchmarked against an Action Chunking Transformer.

Action Chunking Transformer for 7-DOF Pick-and-Place | *PyTorch, MuJoCo*

- 51.6M-parameter CVAE + transformer** imitation policy over a shared ResNet-18 RGBD backbone across three cameras, plus the data engine (randomised MuJoCo environment, teleoperation, **200 episodes** / **~49k frames**); **~73% closed-loop success**.

TECHNICAL SKILLS

Modelling & Techniques: Multimodal & vision-language models (Qwen-VL, FLAVA, VLMO, CLIP, SigLIP), transformers, CVAEs, diffusion (DDPM/DDIM), ConvNeXtV2 / ResNet / U-Net, XLM-RoBERTa, gradient boosting; LoRA & PEFT, knowledge distillation, active learning, conformal prediction, RAG & agentic pipelines, imitation learning

Tools & Infrastructure: Python, C++, SQL, PyTorch, HF Transformers, vLLM, scikit-learn, LangChain, Strands, ROS, MuJoCo, AWS SageMaker (distributed training & real-time inference), Bedrock, Step Functions, Lambda, CDK, CloudWatch